Git Gud — Adaptive Difficulty for Video Games

Balancing game difficulty has been one of the central challenges in video game design since the early days of the medium. Over time, developers have adopted markedly different design philosophies: some allow players to tailor the experience through explicit difficulty settings, others have embraced extreme difficulty as a defining feature of their identity, while yet others make difficulty progression—often under player control—a core gameplay mechanic.

Regardless of the chosen approach, achieving an appropriate balance is crucial to prevent player frustration and disengagement. Players differ widely in their preferences and tolerance for challenge, and an inadequate difficulty balance can severely compromise the overall gameplay experience. Moreover, game difficulty has profound psychological implications, as it plays a key role in inducing a state of flow, in which player engagement and enjoyment are maximized.

To date, no universally satisfactory solution for difficulty balancing has emerged. However, recent advances in artificial intelligence, and large language models (LLMs) in particular, offer promising opportunities to push the state of the art forward. By continuously collecting gameplay data, performing real-time analysis and prediction, and potentially incorporating direct player feedback, it becomes possible to design systems capable of dynamically and continuously adapting game difficulty over time. Such systems aim to sustain an uninterrupted flow experience by aligning challenge levels with the evolving skills and mental state of the player.

The objective of this thesis track is to explore and compare different approaches for implementing dynamic difficulty adjustment (DDA) mechanisms, either within a purpose-built game prototype or through modifications of existing games. The proposed solutions will be empirically evaluated through controlled studies with players, with the goal of assessing their impact on player experience and identifying future research directions. The scope of the thesis is intentionally broad to allow students to tailor the research focus to their interests; prospective students are encouraged to contact Stefano Lambiase for further discussion and guidance.

References: https://consensus.app/search/video-game-difficulty-adaptation/1JNkYvRhTcO4qh-xvMjn-A/?utm_source=share&utm_medium=clipboard

Supervisor: Stefano Lambiase (stla@cs.aau.dk)